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LLM generation novelty through the lens of semantic similarity

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Abstract

Generation novelty is a key indicator of an LLM's ability to generalize, yet measuring it against full pretraining corpora is computationally challenging. Existing evaluations often rely on lexical overlap, failing to detect paraphrased text, or do not consider the full pretraining corpus. We frame novelty as a semantic retrieval problem. This framing enables us to address novelty with modern embedding and indexing pipelines, allowing for efficient analysis at pre-training scale. Specifically, we propose a three-stage framework that retrieves semantically similar samples, reranks them at varying subsequence lengths, and calibrates scores using a human novelty reference for interpretability. We apply this framework to the SmolLM model family and report three key findings: (1) models draw on pretraining data across much longer sequences than previously reported; (2) some task domains systematically promote or suppress generation novelty; and (3) instruction tuning not only alters style but also increases novelty. These results highlight the value of semantic novelty analysis for studying generalization. To support reproducibility and further research, we release ∼20 TB of corpus chunks and index artifacts at https://huggingface.co/datasets/stai-tuebingen/faiss-smollm.

1 Introduction

Large language models (LLMs) now power chatbots, copilots, and autonomous agents with applications across various domains. Their adoption hinges on an implicit assumption: LLMs are capable of generalizing beyond their training data to generate relevant and novel outputs in response to user prompts. Generation novelty provides a useful signal of this capability, indicating whether model outputs extend beyond patterns observed during training. Novelty signals compositional generalization and allows for the assessment of true zero-shot behavior – informing debates about provenance and intellectual property. Thus, measuring a model's generation novelty is a prerequisite for interpreting what LLMs actually learn.

[†]Equal Supervision

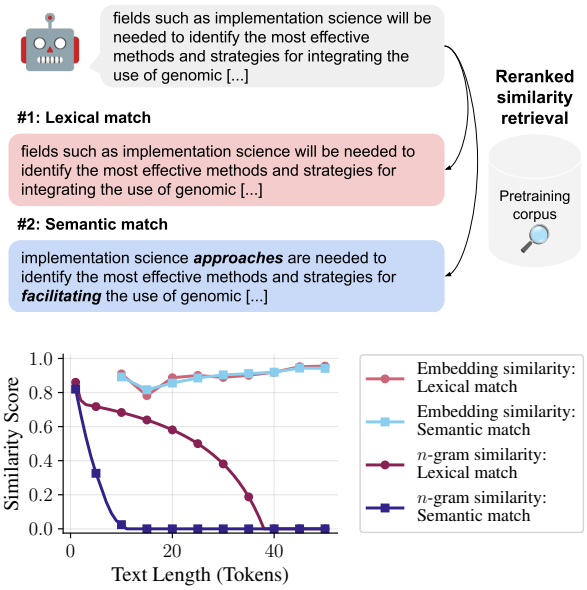

Figure 1: **Efficient retrieval allows for pretraining-scale semantic novelty analysis.** Given a generation, we employ retrieval and reranking pipelines to identify semantically similar samples in the pretraining corpus to analyze generation novelty. Higher similarity indicates lower novelty. While the first-ranked sample is a lexical match, the second-ranked is a semantic match. As text length increases, n-gram overlap drops to zero, falsely suggesting high novelty for paraphrased content. In contrast, embedding similarity correctly identifies the shared meaning, providing a more reliable measure of novelty.

However, analyzing the novelty of LLM generations is a non-trivial task. There is no single, clearly defined notion of novelty, which opens up the question: Is a generation already novel if it does not appear verbatim in the pretraining data? More broadly, what does it mean for a generation to be "reused" from the training data? At the same time, the sheer scale of modern pretraining corpora makes comparison computationally challenging, so that novelty analysis is both a conceptual and a technical problem.

Existing approaches to studying LLM generation novelty mainly address these questions in two directions. One line of work relies on efficient textual overlap metrics (McCoy et al., 2023; Merrill et al., 2024; Padmakumar et al., 2025), which scale well but naturally focus on verbatim reuse and fail to account for paraphrases or stylistic variation. A second line of work investigates novelty within specific domains, such as scientific ideas or biomedical publications (Peng et al., 2025; Ai et al., 2025; Wang et al., 2025c), typically using semantic similarity measured against a curated reference corpus. While effective, these approaches do not account for reuse from other parts of the broader pretraining corpus. Together, these limitations highlight the need for novelty measures that go beyond verbatim overlap at the scale of LLM pretraining corpora.

We address this need by proposing a three-stage framework for analyzing an LLM's generation novelty at scale through the lens of semantic similarity. We argue that a generation should be called novel only if it does not semantically reproduce the training data (cf. Figure 1). From this perspective, "reused" information is not limited to verbatim overlap, but includes paraphrases and reformatting.

Our framework combines semantic retrieval (stage 1) and reranking (stage 2) (Santhanam et al., 2022; Li et al., 2025) to compare LLM generations against their pretraining corpora. However, similarity scores and their relative differences are not directly interpretable as measures of novelty. They are affected by biases in the retrieval pipeline, such as a preference for shorter documents and information located early within documents (Fayyaz et al., 2025; Zhou et al., 2025). Because of this, identical scores can reflect different degrees of novelty, especially when comparing different text lengths or domains. Hence, the third stage of our framework calibrates the raw semantic similarity scores using a baseline of held-out, human-written reference text. By calibrating the scores, we mitigate artifacts of the retrieval pipeline and enable meaningful comparison of scores. Finally, we aggregate these scores into a *novelty profile* which characterizes the generation novelty of a model for a specific dataset and generation text length. This procedure is model- and task-agnostic, and lightweight enough to run on full pretraining corpora. It thus offers a scalable way to assess an LLM's generation novelty.

We demonstrate the utility of our framework through empirical analyses on SmolLM (Allal et al., 2024) and SmolLM2 (Allal et al., 2025), two LLMs with open pretraining corpora. Our analysis reveals unexpected patterns missed by previous lexical methods (McCoy et al., 2023; Merrill et al., 2024). First, both models draw on pretraining data over much longer sequences than previously reported (Merrill et al., 2024). Second, novelty varies systematically by task domain. Third, embedding-based novelty estimates are stable under style shifts from instruction tuning; after accounting for these shifts, instruction tuning substantially increases novelty. These results suggest that instruction tuning shapes not only style but also compositional generation behavior.

Our contributions are as follows:

- We present a semantic similarity–based framework for measuring LLM generation novelty at scale, combining retrieval, reranking, and baseline-calibration to enable comparison against full pretraining corpora while reducing sensitivity to stylistic variation.
- We empirically analyze generation novelty in SmolLM and SmolLM2, uncovering long-range reuse patterns, task-dependent variation, and the impact of instruction tuning beyond stylistic effects.
- We also release the corresponding indices and corpus chunks of SmolLM and SmolLM2 to support replication and extension.

2 Related Work

Novelty measurement in LLMs. Prior work on LLM generation novelty has predominantly relied on textual overlap–based measures, particularly n-gram comparisons. McCoy et al. (2023) introduce n-novelty, defining novelty as the absence of copied text from the training data and quantifying the proportion of non-overlapping n-grams in model outputs. Building on this, Merrill et al. (2024) additionally analyze the probability of generating training n-grams, while Padmakumar et al. (2025) propose novelty as the harmonic mean of n-novelty and generation quality assessed by LLM judges. Wang et al. (2025b) further extend this line of work by introducing task-grams, which capture task-specific n-gram co-occurrences in output and corpus. The task relation introduces a semantic dimension to the n-gram-based analysis. Despite these extensions, all of these approaches rely on

surface-level textual overlap and may treat paraphrases or stylistic variation as novel. We instead define novelty via semantic similarity, enabling analysis beyond surface-form variation. Prior work that emphasizes semantic novelty typically focuses on specific domains, such as scientific ideas or biomedical text (Ai et al., 2025; Peng et al., 2025; Wang et al., 2025c). These approaches also rely on embedding-based similarity, but assess novelty relative to selected reference documents tailored to the target domain. In contrast, our work studies generation novelty with respect to the entire pretraining corpus, independent of the prompt.

Memorization and membership inference. Memorization work (Wu et al., 2025; Feldman and Zhang, 2020) investigates whether specific samples can be elicited verbatim from a model to understand if the model has memorized them. In contrast, our notion of novelty captures whether the underlying *information* is present in the corpus, even if paraphrased. Membership inference attacks (MIA) (Puerto et al., 2025; Mesana et al., 2025; Zhang et al., 2025, 2024) instead ask whether a particular example was part of pretraining, often in adversarial settings. While informative for questions like data privacy, MIAs do not address the broader question of how models generate text that are *not part of* their training data. Our novelty measure, therefore, complements both attribution and memorization/MIA, providing a new perspective on generalization.

Further discussion of prior work relating model generations to pretraining data is provided in Appendix A.

3 Novelty analysis framework

This section presents our framework for studying LLM generation novelty. We introduce the conceptual definition of *semantic* novelty and propose a framework that measures calibrated semantic novelty at LLM pretraining scale.

3.1 Conceptual Definition of Semantic Novelty

We define *generation novelty* as the degree to which an LLM's output expresses information or patterns not readily present in its pretraining corpus. Unlike *lexical novelty*, which is typically measured via n-gram non-overlap (McCoy et al., 2023; Merrill et al., 2024), our definition focuses on *semantic novelty*:

$$\min_{d \in \mathcal{C}} \cos(\phi(y), \phi(d)), y \sim f_\theta(x) \quad (1)$$

where $\mathcal{C}$ is the training corpus, y is the LLM f_θ's output to a prompt x, and ϕ is an embedding function. In other words, semantic novelty is defined as the minimum cosine similarity between a semantic representation of the LLM generation and documents of the pretraining corpus. Under this definition, an output's novelty is determined by its content, rather than its surface form. Hence, this definition serves to distinguish between *reproduction* (generating sequences that exist semantically in the corpus) and *composition* (generating sequences that are internally coherent and correct, yet semantically distinct from any specific training document).

By aggregating semantic signals across a dataset, we can characterize the *novelty profile* of a specific model configuration or training regime. Crucially, we treat novelty not as a binary state, but as a continuous spectrum. We do not claim that a "novel" generation has no relationship to the training data; rather, we measure the extent to which the generated content deviates from the most similar semantic neighbors available in the corpus.

3.2 A Framework for Analysing Semantic Novelty

Our framework operationalizes the measurement of semantic novelty through a model-agnostic, three-stage paradigm applied at the dataset scale. This approach is inspired by standard retrieval-augmented generation (RAG) and information retrieval architectures, where a two-stage process of ranking and re-ranking is the established procedure for balancing search efficiency with semantic precision (Li et al., 2025). Since our framework addresses generation novelty, we further append a third stage that mitigates potential skewness artifacts of the retrieval pipeline, arising from sequence length or domain-specific density of the LLM generations. This stage calibrates the novelty scores with respect to a matched human-level novelty reference so that scores are comparable and interpretable.

The framework aims to characterize the *novelty profile* $\mathcal{N}_\mathcal{Q}(k)$ of a model distribution $\mathcal{Q}$ relative to a domain-matched human distribution $\mathcal{H}$, for each chunk size k. By evaluating across multiple chunk sizes k, we can observe how specific model configurations (e.g., scale or alignment) influence novelty.